Active Brownian Janus particles

Complex disordered systems: Active matter

A Janus colloid of radius $R = 1\,\mu$m swims at $v = 5\,\mu$m/s close to a flat wall, so that its motion is effectively two-dimensional. Its translational and rotational diffusion coefficients are given by the Stokes-Einstein relations

$$D_T = \frac{k_B T}{6\pi\eta R}, \qquad D_r = \frac{k_B T}{8\pi\eta R^3},$$

with $\eta = 10^{-3}$ Pa s and $k_B T \approx 4.1 \times 10^{-21}$ J. The mean squared displacement of an active Brownian particle in two dimensions is

$$\mathrm{MSD}(\tau) = \left(4D_T + 2v^2\tau_R\right)\tau + 2v^2\tau_R^2\left(e^{-\tau/\tau_R} - 1\right), \qquad \tau_R = 1/D_r.$$

1. Expand the MSD for $\tau \ll \tau_R$ and for $\tau \gg \tau_R$, and identify the three regimes discussed in the lectures.
2. Calculate D_T, D_r and τ_R for this particle, and estimate the time τ_1 at which the motion crosses over from diffusive to ballistic.
3. Calculate the long-time effective diffusion coefficient and the Péclet number $\mathrm{Pe} = v/\sqrt{D_T D_r}$. Is this particle strongly active?

Solution

1. For $\tau \ll \tau_R$ we expand the exponential to second order, $e^{-\tau/\tau_R} \simeq 1 - \tau/\tau_R + \tau^2/2\tau_R^2$, and the terms linear in $v^2\tau$ cancel:

   $$\mathrm{MSD}(\tau) \simeq 4D_T\tau + v^2\tau^2.$$

   For $\tau \gg \tau_R$ the exponential vanishes:

   $$\mathrm{MSD}(\tau) \simeq \left(4D_T + 2v^2\tau_R\right)\tau = 4D_{\mathrm{eff}}\,\tau, \qquad D_{\mathrm{eff}} = D_T + \frac{v^2\tau_R}{2}.$$

   So the motion is diffusive at very short times, ballistic once $v^2\tau^2$ takes over, and diffusive again at long times, with a larger diffusion coefficient.
2. Plugging in the numbers

   $$D_T = \frac{k_B T}{6\pi\eta R} \approx 0.22\,\mu\mathrm{m}^2/\mathrm{s}, \qquad D_r = \frac{k_B T}{8\pi\eta R^3} \approx 0.16\,\mathrm{s}^{-1},$$

   so $\tau_R \approx 6$ s. The crossover from diffusive to ballistic motion is where $4D_T\tau_1 = v^2\tau_1^2$:

   $$\tau_1 = \frac{4D_T}{v^2} \approx 0.035\,\mathrm{s},$$

   so the ballistic regime covers more than two decades in time.
3. $$D_{\mathrm{eff}} = D_T + \frac{v^2\tau_R}{2} \approx 0.22 + \frac{25 \times 6.1}{2} \approx 77\,\mu\mathrm{m}^2/\mathrm{s},$$

about 350 times D_T, and

$$\mathrm{Pe} = \frac{v}{\sqrt{D_T D_r}} \approx \frac{5}{\sqrt{0.22 \times 0.16}} \approx 27 \gg 1,$$

so the particle is strongly active.